

Probing Functional Correctness in Diffusion Language Models

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Abstract

Diffusion language models generate text by iteratively denoising all tokens in parallel, but when and where their hidden states encode whether the output will be functionally correct remains unknown. We present the first probing study of diffusion LLM internals, training linear classifiers on hidden states extracted at intermediate denoising steps to predict functional correctness. Across two models (LLaDA-8B, Dream-7B) and four tasks (JSON schema validation, math reasoning, code generation, science QA), we find that (1) correctness signal is already present at step 0, before any denoising occurs, (2) signal emergence is task-dependent, with structural tasks showing flat profiles and reasoning tasks showing gradual buildup, and (3) layer dynamics diverge across architectures. We further demonstrate that per-step probe confidence can identify instances likely to fail, enabling selective generation that avoids 24–98% of wasted compute depending on the task.

1 Introduction

Diffusion language models (dLLMs) have emerged as a promising alternative to autoregressive (AR) generation (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024). Unlike AR models, which generate tokens sequentially, dLLMs produce all output tokens simultaneously through iterative denoising: starting from a fully masked or noisy sequence and progressively refining it over many steps.

This parallel generation process raises a fundamental question: *when* during denoising, and *where* within the model’s layers, does the network encode whether its output will be functionally correct? In AR models, probing classifiers have revealed how linguistic and factual knowledge localizes across layers and positions (Tenney et al., 2019; Geva et al., 2023; Azaria and Mitchell, 2023). However, no analogous study exists for diffusion LLMs, whose bidirectional, iterative generation process

creates a fundamentally different representational landscape.

We address this gap with the first probing study of dLLM hidden states. Our contributions are: (1) we show that correctness signal exists from step 0, exposed simultaneously across all generation positions, a structural advantage of dLLMs over AR models where such signal is concentrated in the final prompt token; (2) we identify task-dependent emergence patterns, with structural tasks decided early and reasoning/code tasks requiring iterative refinement; and (3) we reveal divergent layer dynamics across architectures, with LLaDA concentrating signal in upper layers while Dream migrates signal from upper to lower layers during denoising.

2 Background

Diffusion language models. LLaDA (Nie et al., 2025) uses block masking during training and generates by progressively unmasking tokens over a fixed schedule. Dream (Ye et al., 2025) employs a global masking schedule with a different unmasking strategy. Both models condition on a fully visible prompt and iteratively denoise a generation region of masked tokens.

Probing classifiers. Probing involves training simple classifiers on frozen model representations to test whether specific information is encoded (Blinkov, 2022; Conneau et al., 2018). We adopt this methodology for dLLMs, probing hidden states at intermediate denoising steps.

Related work. Prior studies have probed AR model representations for linguistic structure (Tenney et al., 2019), factual knowledge (Geva et al., 2023), and truthfulness (Azaria and Mitchell, 2023). Confidence-based methods for AR models use early-exit thresholds to skip computation (Schuster et al., 2022; Elbayad et al., 2020). Two concurrent works study dLLM representations from comple-

mentary angles. [Shnaidman et al. \(2025\)](#) demonstrate activation steering for safety behaviors in masked diffusion LLMs, showing that dLLM representations are amenable to linear intervention. [Wang et al. \(2025\)](#) identify temporal oscillation in dLLM outputs, where correct answers appear mid-denoising but are overwritten in later steps. [Mündler et al. \(2025\)](#) propose constrained decoding with context-free grammars to enforce syntactic correctness in dLLM outputs; our work instead asks whether the model’s own representations already encode whether the output will be correct, without external constraints. We complement these prior findings by probing hidden representations for functional correctness across the full layer-step grid, characterizing *when* and *where* correctness signal emerges.

3 Method

Hidden state extraction. Given a dLLM with L layers, we run 128-step denoising and extract hidden states at 7 checkpoint steps $t \in \{0, 1, 4, 16, 32, 64, 127\}$. At each checkpoint, we obtain the full hidden state tensor of shape $(L, \text{gen_length}, d)$. We partition the generation region into 4 equal-length position regions and mean-pool within each region, yielding features of shape $(L, 4, d)$ per instance per step. We choose 4 regions to align with LLaDA’s block length of 32 tokens ($\text{gen_length} / \text{block_length} = 4$ for JSON schema’s 128-token blocks) while keeping the number of features manageable given the limited sample sizes. An ablation over 1, 2, and 4 regions shows best AUC varies by less than 0.02 across configurations (Appendix E).

Probe training. For each (step, layer, region) triple, we reduce dimensionality with PCA to 64 components, apply standard scaling, and train a logistic regression classifier to predict functional correctness ([Pedregosa et al., 2011](#)). We evaluate with 5-fold stratified cross-validation, reporting AUC as our primary metric. Control probes trained on shuffled labels yield $\text{AUC} \approx 0.50$ across all settings, confirming that the observed signal is genuine and not an artifact of the probing pipeline.

Functional correctness. For JSON schema validation, an output is correct if it parses as valid JSON and matches the reference structure. For math reasoning (GSM8K), an output is correct if the extracted numeric answer matches the ground

	JSON	GSM8K	MBPP	ARC
LLaDA-8B	48.5%	66.3%	40.9%	78.2%
Dream-7B	46.0%	61.5%	44.4%	74.5%

Table 1: Baseline functional correctness rates (seed=0, 128 denoising steps).

truth. For code generation (MBPP), an output is correct if the generated function passes all provided test assertions. For science QA (ARC), an output is correct if the extracted answer letter matches the gold label.

Selective generation. We simulate instance-level filtering by training a per-step probe (using the best layer and region from the full analysis). At inference time, if the probe predicts with confidence exceeding a threshold τ that an instance will fail, we skip the remaining denoising steps for that instance entirely, producing no output rather than wasting compute on a generation predicted to be incorrect. We report the fraction of instances filtered and the probe’s classification accuracy.

4 Experimental Setup

Models. We evaluate LLaDA-8B-Instruct (33 layers) ([Nie et al., 2025](#)) and Dream-7B-Instruct (29 layers) ([Ye et al., 2025](#)), two open-weight diffusion LLMs with different masking strategies.

Datasets. We use four tasks that test different capabilities: (1) **JSON schema validation** (272 instances from json-mode-eval-extended), a structural task requiring syntactically and semantically correct JSON output; (2) **GSM8K** (1,319 test instances) ([Cobbe et al., 2021](#)), a multi-step math reasoning task; (3) **MBPP** (257 sanitized test instances) ([Austin et al., 2021b](#)), a code generation task requiring executable Python functions; and (4) **ARC-Challenge** (1,172 test instances) ([Clark et al., 2018](#)), a multiple-choice science reasoning task. Generation lengths are 256, 512, 256, and 256 tokens, respectively.

Baseline functional rates. Table 1 shows the functional correctness rates with 128 denoising steps. All datasets have reasonable class balance, enabling meaningful probe training.

5 Results

5.1 Task-Dependent Emergence

Figure 1 shows AUC heatmaps across layers and denoising steps for all eight model-task combinations. Two distinct emergence patterns are visible.

For **JSON schema**, correctness signal is already strong at step 0 ($\text{AUC} \approx 0.78\text{--}0.80 \pm 0.04$) and remains relatively flat across all denoising steps. This suggests that the prompt encoding alone, before any denoising occurs, already contains sufficient information to predict whether the output will be structurally correct.

For **GSM8K**, step-0 signal is already well above chance ($\text{AUC} \approx 0.67\text{--}0.71 \pm 0.03$), but denoising brings substantially more gain: AUC rises to $\approx 0.78\text{--}0.82 \pm 0.03$ by step 127 (Figure 2).

MBPP follows the same gradual-buildup pattern as GSM8K: step-0 AUC is $\approx 0.75\text{--}0.76 \pm 0.04$ and peaks at $\approx 0.84\text{--}0.87 \pm 0.05$ at later steps (step 64 for LLaDA, step 127 for Dream).

ARC-Challenge shows the weakest and most denoising-dependent signal: step-0 AUC is $\approx 0.61\text{--}0.67$, and signal builds gradually to $\approx 0.71\text{--}0.75$ by step 64–127. Dream’s ARC signal is particularly flat until a sharp jump at the final step ($0.61 \rightarrow 0.71$), suggesting science reasoning requires nearly complete denoising before correctness is predictable.

All four tasks have above-chance step-0 signal, but JSON schema gains only $\Delta\text{AUC} \approx 0.04$ from denoising while GSM8K, MBPP, and ARC gain $\approx 0.08\text{--}0.11$, reflecting the difference between structural and reasoning tasks. Standard deviations across 5-fold CV are ≤ 0.05 for all configurations, indicating stable estimates. Unlike AR models where such signal is concentrated at the final prompt token, dLLMs expose correctness-relevant information simultaneously across all generation positions via bidirectional attention.

Seed reranking at step 1 yields +0.0% improvement for both models, confirming that the probe captures instance-level difficulty rather than seed-specific quality.

5.2 Divergent Layer Dynamics

The heatmaps in Figure 1 reveal strikingly different layer dynamics between the two architectures (full per-step best layers in Table 3, Appendix B).

LLaDA concentrates its best probing signal consistently in upper layers (L22–28) across all denoising steps and all tasks, similar to patterns observed

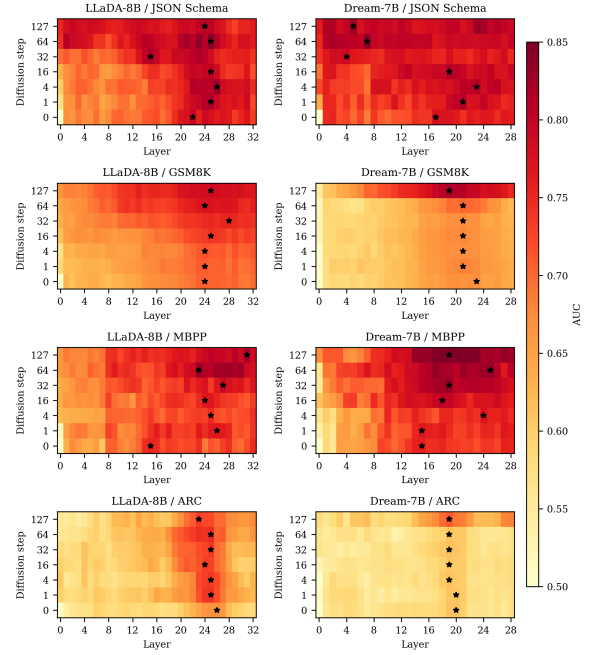


Figure 1: AUC heatmaps (layer $\times$ step) for predicting functional correctness. Stars mark the best layer per step. JSON schema shows strong signal from step 0, while GSM8K, MBPP, and ARC show gradual emergence. Note Dream’s layer migration on JSON schema.

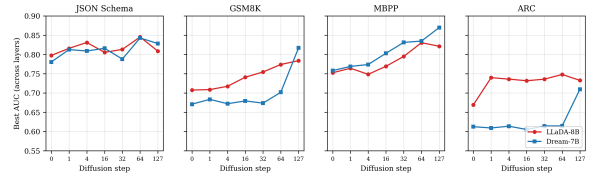


Figure 2: Best AUC across layers at each denoising step. JSON schema is flat (≈ 0.80 from step 0), while GSM8K, MBPP, and ARC show gradual emergence.

in AR transformers.

Dream shows a task-dependent migration pattern. On JSON schema, the best layer migrates from the upper range (L17–23 at steps 0–16) to lower layers (L4–7 at steps 32–127). On GSM8K, MBPP, and ARC, however, Dream’s best layers remain in the mid-to-upper range (L15–25) throughout denoising, with no downward migration. This suggests that the migration occurs only when the task is simple enough for the model to resolve early.

We hypothesize that this difference relates to training origin: LLaDA is trained from scratch, while Dream is fine-tuned from Qwen-2.5 and may inherit its pretrained layer hierarchy, redistributing to lower layers only when the task is simple enough.

	JSON	GSM8K	MBPP	ARC
<i>LLaDA-8B</i>				
Compute saved	98.4%	61.3%	95.5%	82.9%
Clf. accuracy	73.5%	73.5%	66.5%	79.6%
<i>Dream-7B</i>				
Compute saved	96.7%	36.0%	97.8%	61.4%
Clf. accuracy	72.4%	73.5%	70.0%	75.9%

Table 2: Selective generation results ($\tau = 0.80$). Compute saved is the fraction of denoising steps avoided by skipping instances predicted to fail (no output is produced for skipped instances). Clf. acc is the fraction of instances for which the probe correctly predicts functional vs. non-functional.

5.3 Selective Generation

We evaluate whether per-step probe confidence can identify instances whose outputs will be incorrect, enabling the system to skip generation entirely for hopeless instances. Table 2 shows results with a confidence threshold of $\tau = 0.80$.

The results are strongly task-dependent. For JSON schema and MBPP, both models save over 95% of compute: the probe is confident from step 0, identifying most instances predicted to fail immediately. For GSM8K and ARC, savings are more modest (36–83%) because reasoning tasks require more denoising steps before the probe can confidently distinguish hopeless instances.

6 Discussion

Why does step-0 signal exist? At step 0, the dLLM has performed a forward pass over the prompt with masked outputs, producing representations that reflect the model’s “assessment” of the generation task. For structural tasks like JSON generation, this assessment is highly predictive: the model may already encode whether the prompt is one it can handle correctly. For reasoning and code tasks, the prompt assessment provides weaker signal, and the model must iteratively refine its representations through denoising. The dLLM step-0 signal spans all generation positions simultaneously, arising from bidirectional attention over prompt and masked tokens. An AR baseline probe on Qwen-2.5-7B’s last prompt token achieves lower AUC (0.67–0.76) than dLLM step-0 probes (0.61–0.80) on the same instances (Appendix F), confirming that bidirectional attention over masked generation tokens provides signal beyond what AR prompt encoding captures.

Relationship to temporal oscillation. Wang et al. (2025) observe that dLLM outputs can oscillate during denoising: correct answers appear at intermediate steps but are overwritten later. Our probe measures whether hidden states *encode* correctness, not whether the current surface output is correct. On JSON schema, hidden-state signal is flat from step 0 while surface tokens are still being refined, suggesting the model’s internal plan is established early and oscillation reflects surface-level instability rather than changes in the underlying representation of correctness.

Layer dynamics and training origin. LLaDA (trained from scratch) shows stable upper-layer localization resembling AR models, while Dream (fine-tuned from Qwen-2.5) exhibits layer migration on simple tasks but not complex ones. This suggests AR pretraining establishes an upper-layer bias that persists for difficult tasks, but diffusion fine-tuning enables redistribution to lower layers when the task is simple enough.

7 Conclusion

We present the first probing study of diffusion LLM internals, showing that correctness signal exists from step 0 (AUC 0.61–0.80), with task-dependent emergence and divergent layer dynamics across architectures. Selective generation based on probe confidence can avoid 24–98% of wasted compute. An AR baseline comparison and length-stratified controls (AUC 0.72–0.87 after matching) confirm that the signal is genuine and benefits from bidirectional attention over masked tokens.

Limitations

Our probe is a simple logistic regression classifier with PCA-reduced features, which may underestimate the information available in hidden states. Nonlinear probes or probes with access to multiple layers simultaneously might reveal richer structure.

The selective generation analysis is a simulation: we evaluate probe confidence on states extracted during full 128-step generation, but do not validate the approach in an actual inference pipeline where instances predicted to fail would be skipped entirely. Integration would require running the probe at each denoising step and filtering instances in real time, introducing inference overhead that our simulation does not account for.

We study only two models (LLaDA-8B, Dream-7B) and four tasks (JSON schema validation,

GSM8K, MBPP, ARC-Challenge). Broader coverage across more architectures (e.g., MDLM, SEDD) and tasks (e.g., translation, summarization) is needed to establish generality. Both models are in the 7–8B parameter range; it is unclear whether our findings hold for smaller or larger diffusion LLMs.

A control probe trained to predict reference output length (binary median split) achieves higher AUC than the correctness probe across most configurations (Appendix D), indicating that hidden states encode output length more strongly than functional correctness. To control for this confound, we retrain the correctness probe on a length-matched subsample where functional and non-functional instances have identical length distributions. AUC drops modestly but remains well above chance, confirming that the probe captures genuine correctness signal beyond output length (Appendix D). Other surface-level confounders, such as prompt length or the presence of specific formatting tokens, may also inflate probe performance and are not controlled for in this work.

We extract hidden states at only 7 of the 128 denoising steps (log-spaced). This coarse temporal granularity may obscure fine-grained dynamics. For example, Dream-7B’s best layer on JSON schema jumps from layer 19 at step 16 to layer 4 at step 32; without observing the intermediate steps, we cannot determine whether this is a gradual migration or a sudden phase transition.

Finally, probing is inherently correlational: linear separability of a feature does not guarantee that the model actively uses it during generation (Belinkov, 2022). Causal validation through activation interventions, verifying that perturbing the identified representations changes output correctness, remains an important direction for future work.

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A Generation and Probing Details

Denoising procedure. Both models run 128 denoising steps. LLaDA uses block-based unmasking: the generation region is divided into blocks of 32 tokens, and each block is denoised sequentially over $128/(\text{gen_length}/32)$ steps per block. Within each block, the model selects the most confident tokens to unmask at each step. Dream uses a global linear schedule: at step i , the fraction of tokens to unmask is determined by a linearly decreasing timestep schedule from $t = 1$ to $t = \epsilon$, and the most confident masked tokens are unmasked globally.

Prompt construction. For JSON schema, we prepend a system prompt containing the target schema and append “‘ json\n as a generation prefix. For GSM8K, the system prompt instructs the model to solve step-by-step and end with ##### followed by the numeric answer. Both models use their respective chat templates.

Hidden state extraction. At each of the 7 checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$, we run a forward pass with `output_hidden_states=True` and extract the full hidden state tensor across all layers. The generation region is partitioned into 4 equal-length position regions, and we mean-pool within each region, producing a feature vector of dimension d (4096 for LLaDA, 3584 for Dream) per layer per region per instance.

Probe pipeline. For each (step, layer, region) configuration, we construct the feature matrix $X \in R^{n \times d}$ and apply: (1) StandardScaler to zero-mean and unit-variance normalize features, (2) PCA reduction to 64 components, (3) logistic regression with L2 regularization ($C = 1.0$, LBFGS solver, max 1000 iterations). We use 5-fold stratified cross-validation (random state 42) and report the mean AUC across folds.

Computational resources. All generation and feature extraction use NVIDIA A100 (80GB) GPUs on Modal, with 8 parallel A100s per experiment. Models are loaded in bfloat16 precision. Probe training runs on CPU. Total compute: approximately 30 A100-hours across all experiments.

B Best Layer per Step

Table 3: Best layer (by AUC) at each denoising step. Bold values highlight Dream’s migration to lower layers on JSON schema. Other tasks show no such migration.

Step	0	1	4	16	32	64	127
<i>JSON Schema</i>							
LLaDA	22	25	26	25	15	25	24
Dream	17	21	23	19	4	7	5
<i>GSM8K</i>							
LLaDA	24	24	24	25	28	24	25
Dream	23	21	21	21	21	21	19
<i>MBPP</i>							
LLaDA	15	26	25	24	27	23	31
Dream	15	15	24	18	19	25	19
<i>ARC-Challenge</i>							
LLaDA	26	25	25	24	25	25	23
Dream	20	20	19	19	19	19	19

C Selective Generation Details

The selective generation simulation works as follows. For each instance, we iterate through the checkpoint steps in order. At each step, we compute the probe’s confidence as $\max(p, 1 - p)$ where p is the predicted probability of functional correctness. If the probe confidently predicts failure (confidence exceeds threshold τ and $p < 0.5$), we skip the remaining denoising steps for that instance, producing no output. If no checkpoint triggers skipping, the instance proceeds to full generation. Compute savings are computed as $1 - \frac{t_{\text{skip}} + 1}{128}$, where t_{skip} is the step at which filtering occurs.

The main paper reports results at $\tau = 0.80$. Table 4 shows results across a range of thresholds for all configurations.

Table 4: Selective generation results across thresholds (%). “Sv” is compute saved by skipping instances predicted to fail. “Ac” is classification accuracy.

τ	JSON		GSM8K		MBPP		ARC	
	Sv	Ac	Sv	Ac	Sv	Ac	Sv	Ac
<i>LLaDA-8B</i>								
0.60	99.2	64.0	83.4	67.2	99.1	63.4	98.6	78.1
0.70	99.0	69.5	74.2	70.3	98.2	65.8	94.1	79.4
0.80	98.4	73.5	61.3	73.5	95.5	66.5	82.9	79.6
0.90	93.3	75.0	34.1	74.2	90.8	68.1	57.0	79.9
<i>Dream-7B</i>								
0.60	98.8	63.2	85.3	56.9	99.1	72.0	97.2	74.8
0.70	97.9	69.1	66.3	62.5	98.8	70.0	86.1	75.7
0.80	96.7	72.4	36.0	73.5	97.8	70.0	61.4	75.9
0.90	87.3	74.6	6.2	74.8	94.9	73.5	24.2	76.8

D Length Control Probe

We train a control probe (same pipeline) to predict binary output length (above/below median) instead of correctness. To control for the confound, we re-train the correctness probe on a length-matched subsample where functional and non-functional instances have identical length distributions (binned matching, 10 quantile bins).

Table 5: Best AUC for correctness, length control, and length-matched correctness probes. After controlling for length, correctness AUC drops modestly but remains well above chance.

	Corr.	Length	Corr. (matched)
<i>JSON Schema</i>			
LLaDA	0.85	0.98	0.81
Dream	0.84	0.99	0.75
<i>GSM8K</i>			
LLaDA	0.79	0.86	0.75
Dream	0.82	0.87	0.80
<i>MBPP</i>			
LLaDA	0.84	0.78	0.84
Dream	0.87	0.80	0.87
<i>ARC</i>			
LLaDA	0.75	0.96	0.75
Dream	0.71	0.98	0.72

E Spatial Pooling Ablation

We pool the 4 extracted regions into 1 (global average), 2 (pairwise average), or 4 (concatenation) groups and re-run the probe.

Table 6: Best AUC by number of position regions. Spread is less than 0.03 within each configuration.

	1 region	2 regions	4 regions
<i>JSON Schema</i>			
LLaDA	0.848	0.854	0.840
Dream	0.846	0.828	0.847
<i>GSM8K</i>			
LLaDA	0.784	0.788	0.784
Dream	0.818	0.814	0.812
<i>MBPP</i>			
LLaDA	0.842	0.860	0.834
Dream	0.864	0.868	0.856
<i>ARC-Challenge</i>			
LLaDA	0.747	0.740	0.744
Dream	0.710	0.711	0.716

F AR Baseline Probe

We probe Qwen-2.5-7B-Instruct, the AR base model of Dream, using the last prompt token hidden state (one forward pass, no generation). Sub-A uses dLLM functional correctness labels; Sub-B generates with Qwen (greedy) and uses its own correctness labels.

Table 7: AR vs. dLLM step-0 probe AUC. The AR probe uses only the last prompt token; dLLM probes use mean-pooled generation region hidden states at step 0. Sub-A predicts dLLM labels; Sub-B predicts AR’s own labels.

	JSON	GSM8K	MBPP	ARC
Qwen Sub-A (dLLM)	0.74	0.69	0.76	0.67
Qwen Sub-B (AR)	0.77	0.77	0.69	0.74
LLaDA step-0	0.78	0.71	0.76	0.67
Dream step-0	0.80	0.71	0.76	0.61